

Privacy-Aware Continual Knowledge Distillation for Cross-Domain Medical Image Classification

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Motivation and Objective

Models used in clinical settings have to be updated periodically owing to the development of new diagnostic tasks and medical imaging techniques. Retraining using all the past data is difficult due to privacy restrictions and space constraints that do not allow accessing previous patient data. Training a new model to learn a new task without having access to previous data may result in catastrophic forgetting, resulting in poor performance on previous tasks. To solve the problem, we leverage **Continual Learning (CL)** and **Knowledge Distillation (KD)** in order to make a compact deployable model capable of learning a series of medical imaging tasks without forgetting the previous knowledge. In addition, it is worth noting that continual learning by its nature does not assume access to past data, which poses privacy issues as well since it is possible to get some information about the persons whose data was used for training the model. We therefore add a privacy analysis to measure and reduce this leakage.

Proposed Methodology

The ResNet-18 teacher model is first trained on Task A and distilled into a light-weight MobileNetV3 student using logit distillation. It has a shared feature extractor backbone and task-specific classifier heads. The distilled student model is then fine-tuned on Task B incrementally without access to any Task A data through known methods of continual learning:

- Regularization-based: **Elastic Weight Consolidation (EWC)**
- Distillation-based: **Learning without Forgetting (LwF)**
- Architectural stabilization: **BatchNorm freezing**

Our work introduces a new method of assessing the privacy issues associated with training the model using Membership Inference Attack (MIA), where we try to measure the leakage of information from the model about its training dataset. Differential Privacy (DP-SGD) has been used to overcome this issue and study the utility-privacy trade-off over different privacy budgets ϵ .

Dataset and Setup

- Task A — **Chest X-Ray Pneumonia** (2 classes: Normal, Pneumonia)
- Task B — **Brain Tumor MRI** (4 classes: glioma, meningioma, no-tumor, pituitary)

The tasks come from two different imaging modalities, X-ray and MRI, resulting in an evident domain shift, which corresponds to the environment where catastrophic forgetting happens the most frequently. This task comes in a sequence of two tasks continual learning setup; a sample of data for each dataset is selected appropriately to keep the training feasible.

Expected Contributions

- PyTorch-based architecture that combines knowledge distillation, continual learning, and privacy modules.
- Baselines comparison: naive fine-tuning (lower bound) vs EWC, LwF, and BatchNorm freezing in terms of Task-A retention and Task-B performance.
- Ablations that explore how different hyperparameters like λ (EWC penalty) and BatchNorm freezing impact our results.
- An entirely new privacy analysis on membership inference leakage and differential privacy mechanism (DP-SGD), privacy-utility plots.
- Final paper in the CVPR 2025 template with a link to a public code repository.